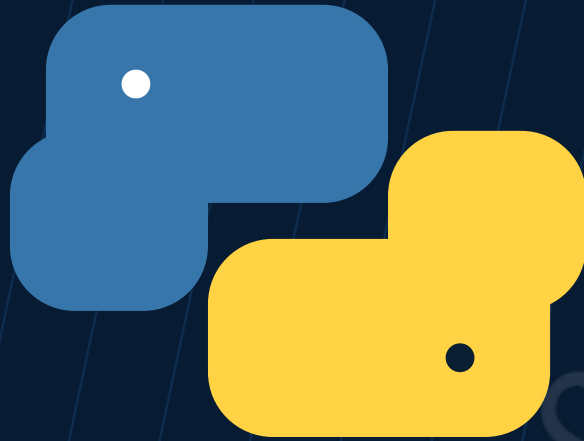




100+ Programs to Master Python

Code Snippets with Output for Practice and Interviews

```
Python

print("Practice Python Daily")
# Output: Practice Python Daily
```

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Practice - Learn - Prepare

1. Hello World Program



Python Code

```
1 print("Hello, World!")
```

Output

Hello, World!

2. Add Two Numbers



Python Code

```
1 a = 5
2 b = 7
3 sum_result = a + b
4 print("Sum:", sum_result)
```

Output

Sum: 12

3. Simple Calculator



Python Code

```
1 a = 20
2 b = 5
3 print("Addition:", a + b)
4 print("Subtraction:", a - b)
5 print("Multiplication:", a * b)
6 print("Division:", a / b)
```

Output

Addition: 25
Subtraction: 15
Multiplication: 100
Division: 4.0

4. Swap Two Variables



Python Code

```
1 a = 10
2 b = 20
3 a, b = b, a
4 print("a =", a)
5 print("b =", b)
```

Output

```
a = 20
b = 10
```

5. Check Even or Odd



Python Code

```
1 num = 9
2 if num % 2 == 0:
3     print("Even number")
4 else:
5     print("Odd number")
```

Output

```
Odd number
```

6. Check Positive, Negative or Zero



Python Code

```
1 num = -4
2 if num > 0:
3     print("Positive")
4 elif num == 0:
5     print("Zero")
6 else:
7     print("Negative")
```

Output

```
Negative
```

7. Find Largest of Three Numbers



Python Code

```
1 a, b, c = 12, 45, 27
2 largest = max(a, b, c)
3 print("Largest number:", largest)
```

Output

Largest number: 45

8. Area of a Circle



Python Code

```
1 import math
2 radius = 5
3 area = math.pi * radius * radius
4 print("Area:", round(area, 2))
```

Output

Area: 78.54

9. Area of a Triangle



Python Code

```
1 base = 10
2 height = 6
3 area = 0.5 * base * height
4 print("Area:", area)
```

Output

Area: 30.0

10. Convert Celsius to Fahrenheit



Python Code

```
1 celsius = 37
2 fahrenheit = (celsius * 9/5) + 32
3 print("Fahrenheit:", fahrenheit)
```

Output

Fahrenheit: 98.6

11. Convert Kilometers to Miles



Python Code

```
1 kilometers = 10
2 miles = kilometers * 0.621371
3 print("Miles:", round(miles, 2))
```

Output

Miles: 6.21

12. Find Square Root



Python Code

```
1 import math
2 num = 64
3 print("Square root:", math.sqrt(num))
```

Output

Square root: 8.0

13. Generate Random Number



Python Code

```
1 import random
2 random.seed(1)
3 print("Random number:", random.randint(1, 100))
```

Output

Random number: 18

14. Check Leap Year



Python Code

```
1 year = 2024
2 if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
3     print("Leap year")
4 else:
5     print("Not a leap year")
```

Output

Leap year

15. Find Factorial



Python Code

```
1 num = 5
2 factorial = 1
3 for i in range(1, num + 1):
4     factorial *= i
5 print("Factorial:", factorial)
```

Output

Factorial: 120

16. Factorial Using Recursion



Python Code

```
1 def factorial(n):
2     if n == 0:
3         return 1
4     return n * factorial(n - 1)
5
6 print(factorial(5))
```

Output

120

17. Check Prime Number



Python Code

```
1 num = 29
2 is_prime = True
3 for i in range(2, int(num ** 0.5) + 1):
4     if num % i == 0:
5         is_prime = False
6         break
7 print("Prime" if is_prime else "Not prime")
```

Output

Prime

18. Print Prime Numbers in Range



Python Code

```
1  for num in range(2, 11):
2      for i in range(2, num):
3          if num % i == 0:
4              break
5      else:
6          print(num)
```

Output

```
2
3
5
7
```

19. Multiplication Table



Python Code

```
1  num = 5
2  for i in range(1, 6):
3      print(f"{num} x {i} = {num * i}")
```

Output

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
```

20. Fibonacci Series



Python Code

```
1  n = 7
2  a, b = 0, 1
3  for _ in range(n):
4      print(a)
5      a, b = b, a + b
```

Output

```
0
1
1
2
3
5
8
```

21. Fibonacci Using Recursion



Python Code

```
1  def fib(n):
2      if n <= 1:
3          return n
4      return fib(n - 1) + fib(n - 2)
5
6  for i in range(6):
7      print(fib(i))
```

Output

```
0
1
1
2
3
5
```

22. Sum of Natural Numbers



Python Code

```
1 n = 10
2 total = sum(range(1, n + 1))
3 print("Sum:", total)
```

Output

Sum: 55

23. Cube Sum of Natural Numbers



Python Code

```
1 n = 4
2 total = sum(i ** 3 for i in range(1, n + 1))
3 print("Cube sum:", total)
```

Output

Cube sum: 100

24. Check Armstrong Number



Python Code

```
1 num = 153
2 power = len(str(num))
3 total = sum(int(digit) ** power for digit in str(num))
4 print("Armstrong" if total == num else "Not Armstrong")
```

Output

Armstrong

25. Find Armstrong Numbers in Range

Python Code

```
1  for num in range(100, 500):
2      power = len(str(num))
3      total = sum(int(d) ** power for d in str(num))
4      if total == num:
5          print(num)
```

Output

153
370
371
407

26. Reverse a Number

Python Code

```
1  num = 12345
2  reverse = int(str(num)[::-1])
3  print("Reverse:", reverse)
```

Output

Reverse: 54321

27. Check Palindrome Number

Python Code

```
1  num = 121
2  text = str(num)
3  print("Palindrome" if text == text[::-1] else "Not palindrome")
```

Output

Palindrome

28. Find GCD

Python Code

```
1 import math
2 a, b = 48, 18
3 print("GCD:", math.gcd(a, b))
```

Output

GCD: 6

29. Find LCM

Python Code

```
1 import math
2 a, b = 12, 18
3 lcm = abs(a * b) // math.gcd(a, b)
4 print("LCM:", lcm)
```

Output

LCM: 36

30. Decimal to Binary, Octal and Hexadecimal

Python Code

```
1 num = 27
2 print("Binary:", bin(num))
3 print("Octal:", oct(num))
4 print("Hexadecimal:", hex(num))
```

Output

Binary: 0b11011
Octal: 0o33
Hexadecimal: 0x1b

31. Find ASCII Value



Python Code

```
1 char = "A"
2 print("ASCII value:", ord(char))
```

Output

ASCII value: 65

32. Check Vowel or Consonant



Python Code

```
1 char = "e"
2 if char.lower() in "aeiou":
3     print("Vowel")
4 else:
5     print("Consonant")
```

Output

Vowel

33. Count Vowels in String



Python Code

```
1 text = "Python Programming"
2 count = sum(1 for ch in text.lower() if ch in "aeiou")
3 print("Vowels:", count)
```

Output

Vowels: 4

34. Remove Punctuation



Python Code

```
1 import string
2 text = "Hello, World!"
3 clean = "".join(ch for ch in text if ch not in string.punctuation)
4 print(clean)
```

Output

Hello World

35. Sort Words Alphabetically



Python Code

```
1 text = "banana apple cherry"
2 words = text.split()
3 words.sort()
4 print(words)
```

Output

['apple', 'banana', 'cherry']

36. Check Anagram



Python Code

```
1 s1 = "listen"
2 s2 = "silent"
3 print("Anagram" if sorted(s1) == sorted(s2) else "Not anagram")
```

Output

Anagram

37. String Palindrome



Python Code

```
1 text = "madam"
2 print("Palindrome" if text == text[::-1] else "Not palindrome")
```

Output

Palindrome

38. Count Words in String



Python Code

```
1 text = "Python is easy to learn"
2 words = text.split()
3 print("Words:", len(words))
```

Output

Words: 5

39. Reverse a String



Python Code

```
1 text = "Python"
2 print(text[::-1])
```

Output

nohtyP

40. Change String Case



Python Code

```
1 text = "Python"
2 print(text.upper())
3 print(text.lower())
```

Output

```
PYTHON
python
```

41. Find Length of String



Python Code

```
1 text = "Interview"
2 print("Length:", len(text))
```

Output

```
Length: 9
```

42. Replace Vowels in String



Python Code

```
1 text = "education"
2 for vowel in "aeiou":
3     text = text.replace(vowel, "*")
4 print(text)
```

Output

```
*d*c*t**n
```

43. Check Binary String



Python Code

```
1 text = "101010"
2 print("Binary string" if set(text) <= {"0", "1"} else "Not binary")
```

Output

Binary string

44. Find Duplicate Characters



Python Code

```
1 text = "programming"
2 duplicates = sorted({ch for ch in text if text.count(ch) > 1})
3 print(duplicates)
```

Output

['g', 'm', 'r']

45. List Sum



Python Code

```
1 numbers = [10, 20, 30]
2 print("Sum:", sum(numbers))
```

Output

Sum: 60

46. List Product



Python Code

```
1 numbers = [2, 3, 4]
2 product = 1
3 for n in numbers:
4     product *= n
5 print("Product:", product)
```

Output

Product: 24

47. Largest Number in List



Python Code

```
1 numbers = [12, 45, 7, 89, 34]
2 print("Largest:", max(numbers))
```

Output

Largest: 89

48. Smallest Number in List



Python Code

```
1 numbers = [12, 45, 7, 89, 34]
2 print("Smallest:", min(numbers))
```

Output

Smallest: 7

49. Second Largest Number



Python Code

```
1 numbers = [12, 45, 7, 89, 34]
2 unique = sorted(set(numbers), reverse=True)
3 print("Second largest:", unique[1])
```

Output

Second largest: 45

50. Even Numbers in List



Python Code

```
1 numbers = [1, 2, 3, 4, 5, 6]
2 evens = [n for n in numbers if n % 2 == 0]
3 print(evens)
```

Output

[2, 4, 6]

51. Odd Numbers in List



Python Code

```
1 numbers = [1, 2, 3, 4, 5, 6]
2 odds = [n for n in numbers if n % 2 != 0]
3 print(odds)
```

Output

[1, 3, 5]

52. Remove Empty Lists



Python Code

```
1 items = [[1, 2], [], [3], [], [4, 5]]
2 result = [item for item in items if item]
3 print(result)
```

Output

```
[[1, 2], [3], [4, 5]]
```

53. Copy a List



Python Code

```
1 original = [1, 2, 3]
2 copy_list = original.copy()
3 print(copy_list)
```

Output

```
[1, 2, 3]
```

54. Count Occurrences in List



Python Code

```
1 numbers = [1, 2, 2, 3, 2, 4]
2 print("Count of 2:", numbers.count(2))
```

Output

```
Count of 2: 3
```

55. Remove Duplicates from List



Python Code

```
1 numbers = [1, 2, 2, 3, 3, 4]
2 result = list(dict.fromkeys(numbers))
3 print(result)
```

Output

```
[1, 2, 3, 4]
```

56. List Comprehension Squares



Python Code

```
1 squares = [n ** 2 for n in range(1, 6)]
2 print(squares)
```

Output

```
[1, 4, 9, 16, 25]
```

57. Merge Two Lists



Python Code

```
1 a = [1, 3, 5]
2 b = [2, 4, 6]
3 merged = sorted(a + b)
4 print(merged)
```

Output

```
[1, 2, 3, 4, 5, 6]
```

58. Rotate List



Python Code

```
1 numbers = [1, 2, 3, 4, 5]
2 k = 2
3 rotated = numbers[k:] + numbers[:k]
4 print(rotated)
```

Output

```
[3, 4, 5, 1, 2]
```

59. Check Monotonic List



Python Code

```
1 numbers = [1, 2, 2, 3]
2 inc = all(numbers[i] <= numbers[i+1] for i in range(len(numbers)-1))
3 dec = all(numbers[i] >= numbers[i+1] for i in range(len(numbers)-1))
4 print(inc or dec)
```

Output

```
True
```

60. Add Two Matrices



Python Code

```
1 a = [[1, 2], [3, 4]]
2 b = [[5, 6], [7, 8]]
3 result = [[a[i][j] + b[i][j] for j in range(2)] for i in range(2)]
4 print(result)
```

Output

```
[[6, 8], [10, 12]]
```

61. Transpose a Matrix



Python Code

```
1 matrix = [[1, 2, 3], [4, 5, 6]]
2 transpose = [[row[i] for row in matrix] for i in range(3)]
3 print(transpose)
```

Output

```
[[1, 4], [2, 5], [3, 6]]
```

62. Multiply Two Matrices



Python Code

```
1 a = [[1, 2], [3, 4]]
2 b = [[5, 6], [7, 8]]
3 result = [[sum(a[i][k] * b[k][j] for k in range(2)) for j in range(2)] for i
4             in range(2)]
5 print(result)
```

Output

```
[[19, 22], [43, 50]]
```

63. Dictionary Sum of Values



Python Code

```
1 data = {"a": 10, "b": 20, "c": 30}
2 print("Sum:", sum(data.values()))
```

Output

```
Sum: 60
```

64. Merge Dictionaries



Python Code

```
1 a = {"x": 1, "y": 2}
2 b = {"z": 3}
3 merged = {**a, **b}
4 print(merged)
```

Output

```
{'x': 1, 'y': 2, 'z': 3}
```

65. Sort Dictionary by Key



Python Code

```
1 data = {"banana": 2, "apple": 5, "cherry": 1}
2 result = dict(sorted(data.items()))
3 print(result)
```

Output

```
{'apple': 5, 'banana': 2, 'cherry': 1}
```

66. Sort Dictionary by Value



Python Code

```
1 data = {"a": 3, "b": 1, "c": 2}
2 result = dict(sorted(data.items(), key=lambda item: item[1]))
3 print(result)
```

Output

```
{'b': 1, 'c': 2, 'a': 3}
```

67. Extract Unique Dictionary Values



Python Code

```
1 data = {"a": 10, "b": 20, "c": 10}
2 unique = list(set(data.values()))
3 print(unique)
```

Output

```
[10, 20]
```

68. Convert Dictionary to List of Tuples



Python Code

```
1 data = {"B": 2, "A": 1, "C": 3}
2 result = sorted(data.items())
3 print(result)
```

Output

```
[('A', 1), ('B', 2), ('C', 3)]
```

69. Create Mapping Lowercase to Uppercase



Python Code

```
1 letters = ["a", "b", "c"]
2 mapping = {letter: letter.upper() for letter in letters}
3 print(mapping)
```

Output

```
{'a': 'A', 'b': 'B', 'c': 'C'}
```

70. Check Key Exists in Dictionary



Python Code

```
1 data = {"name": "Ali", "age": 22}
2 print("age" in data)
```

Output

True

71. Binary Search Algorithm



Python Code

```
1 def binary_search(arr, target):
2     left, right = 0, len(arr) - 1
3     while left <= right:
4         mid = (left + right) // 2
5         if arr[mid] == target:
6             return mid
7         elif arr[mid] < target:
8             left = mid + 1
9         else:
10            right = mid - 1
11    return -1
12
13 print(binary_search([1, 3, 5, 7], 5))
```

Output

2

72. Linear Search Algorithm



Python Code

```
1 arr = [4, 2, 7, 9]
2 target = 7
3 for index, value in enumerate(arr):
4     if value == target:
5         print("Found at index", index)
6         break
```

Output

Found at index 2

73. Bubble Sort Algorithm



Python Code

```
1 arr = [5, 1, 4, 2]
2 for i in range(len(arr)):
3     for j in range(0, len(arr) - i - 1):
4         if arr[j] > arr[j + 1]:
5             arr[j], arr[j + 1] = arr[j + 1], arr[j]
6 print(arr)
```

Output

[1, 2, 4, 5]

74. Selection Sort Algorithm



Python Code

```
1 arr = [64, 25, 12, 22]
2 for i in range(len(arr)):
3     min_i = i
4     for j in range(i + 1, len(arr)):
5         if arr[j] < arr[min_i]:
6             min_i = j
7     arr[i], arr[min_i] = arr[min_i], arr[i]
8 print(arr)
```

Output

[12, 22, 25, 64]

75. Insertion Sort Algorithm

Python Code

```
1  arr = [12, 11, 13, 5]
2  for i in range(1, len(arr)):
3      key = arr[i]
4      j = i - 1
5      while j >= 0 and key < arr[j]:
6          arr[j + 1] = arr[j]
7          j -= 1
8      arr[j + 1] = key
9  print(arr)
```

Output

[5, 11, 12, 13]

76. Simple Stack Implementation

Python Code

```
1  stack = []
2  stack.append(10)
3  stack.append(20)
4  print("Popped:", stack.pop())
5  print("Stack:", stack)
```

Output

Popped: 20
Stack: [10]

77. Simple Queue Implementation

Python Code

```
1 from collections import deque
2 queue = deque()
3 queue.append(10)
4 queue.append(20)
5 print("Removed:", queue.popleft())
6 print("Queue:", list(queue))
```

Output

Removed: 10
Queue: [20]

78. Class and Object

Python Code

```
1 class Student:
2     def __init__(self, name):
3         self.name = name
4
5 student = Student("Ali")
6 print("Student:", student.name)
```

Output

Student: Ali

79. Inheritance Example



Python Code

```
1 class Animal:
2     def speak(self):
3         return "Animal sound"
4
5 class Dog(Animal):
6     def speak(self):
7         return "Bark"
8
9 print(Dog().speak())
```

Output

Bark

80. Calculate BMI



Python Code

```
1 weight = 70
2 height = 1.75
3 bmi = weight / (height ** 2)
4 print("BMI:", round(bmi, 2))
```

Output

BMI: 22.86

81. Compound Interest



Python Code

```
1 p = 1000
2 r = 0.05
3 t = 2
4 a = p * (1 + r) ** t
5 print("Amount:", round(a, 2))
```

Output

Amount: 1102.5

82. Simple Interest



Python Code

```
1 p = 1000
2 r = 5
3 t = 2
4 interest = (p * r * t) / 100
5 print("Interest:", interest)
```

Output

Interest: 100.0

83. Find Mean of Numbers



Python Code

```
1 numbers = [10, 20, 30]
2 mean = sum(numbers) / len(numbers)
3 print("Mean:", mean)
```

Output

Mean: 20.0

84. Find Median of Numbers



Python Code

```
1 numbers = [5, 1, 9, 3, 7]
2 numbers.sort()
3 median = numbers[len(numbers)//2]
4 print("Median:", median)
```

Output

Median: 5

85. Find Mode of Numbers



Python Code

```
1 numbers = [1, 2, 2, 3, 3, 3]
2 mode = max(set(numbers), key=numbers.count)
3 print("Mode:", mode)
```

Output

Mode: 3

86. Read and Write File



Python Code

```
1 with open("sample.txt", "w") as file:
2     file.write("Hello File")
3
4 with open("sample.txt", "r") as file:
5     print(file.read())
```

Output

Hello File

87. Count Lines in File



Python Code

```
1 text = "Line 1\nLine 2\nLine 3"
2 lines = text.splitlines()
3 print("Total lines:", len(lines))
```

Output

Total lines: 3

88. Handle Division by Zero



Python Code

```
1 try:
2     result = 10 / 0
3 except ZeroDivisionError:
4     print("Cannot divide by zero")
```

Output

Cannot divide by zero

89. Use Lambda Function



Python Code

```
1 square = lambda x: x * x
2 print(square(6))
```

Output

36

90. Use Map Function



Python Code

```
1 numbers = [1, 2, 3]
2 squares = list(map(lambda x: x * x, numbers))
3 print(squares)
```

Output

[1, 4, 9]

91. Use Filter Function



Python Code

```
1 numbers = [1, 2, 3, 4, 5]
2 evens = list(filter(lambda x: x % 2 == 0, numbers))
3 print(evens)
```

Output

```
[2, 4]
```

92. Use Reduce Function



Python Code

```
1 from functools import reduce
2 numbers = [1, 2, 3, 4]
3 product = reduce(lambda a, b: a * b, numbers)
4 print(product)
```

Output

```
24
```

93. Generate Calendar



Python Code

```
1 import calendar
2 print(calendar.month(2024, 1))
```

Output

```
January 2024 calendar is displayed
```

94. Current Date and Time



Python Code

```
1 from datetime import datetime
2 now = datetime(2024, 1, 1, 10, 30)
3 print(now.strftime("%Y-%m-%d %H:%M"))
```

Output

2024-01-01 10:30

95. Calculate Age



Python Code

```
1 birth_year = 2000
2 current_year = 2024
3 age = current_year - birth_year
4 print("Age:", age)
```

Output

Age: 24

96. Password Validation



Python Code

```
1 password = "Python@123"
2 has_upper = any(ch.isupper() for ch in password)
3 has_digit = any(ch.isdigit() for ch in password)
4 has_symbol = any(not ch.isalnum() for ch in password)
5 print(has_upper and has_digit and has_symbol)
```

Output

True

97. Email Username Extractor



Python Code

```
1 email = "student@example.com"
2 username = email.split("@")[0]
3 print(username)
```

Output

student

98. Find Common Elements



Python Code

```
1 a = [1, 2, 3, 4]
2 b = [3, 4, 5, 6]
3 common = list(set(a) & set(b))
4 print(common)
```

Output

[3, 4]

99. Find Missing Number



Python Code

```
1 numbers = [1, 2, 4, 5]
2 n = 5
3 missing = sum(range(1, n + 1)) - sum(numbers)
4 print("Missing:", missing)
```

Output

Missing: 3

100. Check Isogram



Python Code

```
1 word = "python"
2 print(len(word) == len(set(word)))
```

Output

True